

Wonjun Lee

PH.D. CANDIDATE IN ELECTRICAL ENGINEERING, CILAB, KAIST

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I am currently pursuing an Ph.D. degree in Electrical Engineering at KAIST. My academic journey began at Gyeonggi Science High School for the Gifted, which fostered my passion for research and innovation. I then earned my B.S. in Electrical Engineering and Computer Science at KAIST, and M.S. in Electrical Engineering at KAIST. I enjoy engaging with creative challenges and value building meaningful relationships along the way. Please feel free to explore my academic background below and contact me directly for further details.

Research Interest

As AI's role expands across various fields, building **trustworthy AI** is crucial. I am particularly interested in AI security, as vulnerabilities compromise reliability and fairness.

Additionally, as humans rely on various senses for judgment, I focus on research in **Multi-Modal Learning** and **AI Security**. My goal is to help bridge the understanding between AI and humans. My work explores the following, but not limited to:

- **Multi-Modal Learning**: multi-modal large language models, efficient multi-modal learning
- **AI Security**: adversarial attack, jailbreaking and their defense
- **Multimedia Forensics**: detecting forgeries, synthetic images, and deepfakes

Education

Korea Advanced Institute of Science and Technology (KAIST)

PH.D. IN ELECTRICAL ENGINEERING

Daejeon, S.Korea

Mar. 2025 - Present

Korea Advanced Institute of Science and Technology (KAIST)

M.S. IN ELECTRICAL ENGINEERING

Daejeon, S.Korea

Mar. 2023 - Feb. 2025

Korea Advanced Institute of Science and Technology (KAIST)

B.S. IN ELECTRICAL ENGINEERING & SCHOOL OF COMPUTING (DOUBLE MAJOR)

Daejeon, S.Korea

Mar. 2019 - Feb. 2023

Gyeonggi Science High School for the Gifted (GSHS)

MAJOR IN MATH, PHYSICS, AND BIOLOGY

Suwon, Gyeonggi-do, S.Korea

Mar. 2016 - Feb. 2019

Publication

INTERNATIONAL

Jailbreak to Protect: Buffering Harmful Fine-Tuning via Temporary Jailbreaking LoRA in Large Language Models

2026

INTERNATIONAL CONFERENCE ON MACHINE LEARNING (ICML) **SPOTLIGHT (TOP 2.2%)**

Multimodal LLM Safety

Seokil Ham, Jaehyuk Jang, **Wonjun Lee**, Changick Kim

Generalizable Prompt Tuning for Audio-Language Models via Semantic Expansion

2026

THE 64TH ANNUAL MEETING OF THE ASSOCIATION FOR COMPUTATIONAL LINGUISTICS (ACL FINDINGS)

ALM Prompt Tuning

Jaehyuk Jang*, **Wonjun Lee***, Kangwook Ko*, Changick Kim, Changick Kim (*: Equal Contribution)

Efficient Test-Time Optimization for Depth Completion via Low-Rank Decoder Adaptation

2026

PREPRINT

Depth Estimation, VFM

Minseok Seo*, **Wonjun Lee***, Jaehyuk Jang, Changick Kim (*: Equal Contribution)

SELF: Selective Fusion of Identity for Generalizable Deepfake Detection

2025

PREPRINT

Deepfake Detection

Younghun Kim, Minsuk Jang, Myung-Joon Kwon, **Wonjun Lee**, Changick Kim

Benign-to-Toxic Jailbreaking: Inducing Harmful Responses from Harmless Prompts

2025

PREPRINT

Multimodal LLM Safety

Hee-Seon Kim, Minbeom Kim, **Wonjun Lee**, Kihyun Kim, Changick Kim

SAFIRE: Segment Any Forged Image Region

2025

AAAI CONFERENCE ON ARTIFICIAL INTELLIGENCE (AAAI)

Image Forgery Localization

Myung-Joon Kwon*, **Wonjun Lee***, Seung-Hun Nam, Minji Son, Changick Kim (*: Equal Contribution)

FRIDAY: Mitigating Unintentional Facial Identity in Deepfake Detectors Guided by Facial Recognizers

2024

VISUAL COMMUNICATIONS AND IMAGE PROCESSING (VCIP) **ORAL PRESENTATION**

Deepfake Detection

Younghun Kim, Myung-Joon Kwon, **Wonjun Lee**, Changick Kim

DOMESTIC

Image-Text Order Matters: Input Ordering Effects in LVLM Jailbreaks

2026

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE) **LG ELECTRONICS PAPER AWARD**

Multimodal LLM Safety

Kihyun Seol, **Wonjun Lee**, Kihyun Kim, Minbeom Kim, Changick Kim

Safety Analysis of Image Verification-Based Reasoning Methods in MLLMs

2026

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE)

Multimodal LLM Safety

Jihyun Lee, **Wonjun Lee**, Kihyun Kim, Hee-Seon Kim, Changick Kim

Image-Based Universal Jailbreak Attack via Safety Boundary Approximation

2025

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE)

Multimodal LLM Safety

Kihyun Kim, Hee-Seon Kim, Minbeom Kim, **Wonjun Lee**, Changick Kim

Ensembling Pretrained Models for Enhanced Generalization in Synthetic Image Detection

2024

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE) **LG ELECTRONICS PAPER AWARD**

Synthetic Image Detection

Myung-Joon Kwon, **Wonjun Lee**, Younghun Kim, Changick Kim

Combating Bias: Corrective Measures Through Adversarial Training in Neural Networks

2024

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE) **LG ELECTRONICS PAPER AWARD**

De-Biasing, Adversarial Training

Wonjun Lee, Myung-Joon Kwon, Younghun Kim, Changick Kim

Improving the Generalization Performance of Deepfake Detection Through a Partial Image Manipulation Augmentation Technique

2024

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE)

Deepfake Detection

Younghun Kim, Myung-Joon Kwon, **Wonjun Lee**, Changick Kim

Research Experience

Roen Surgical

Daejeon, S.Korea

KAIST EE EXTERNSHIP PROGRAM SECOND COHORT

Jun.2022 – Dec.2022

SAMSUNG Electronics CE/IM Division Mobile Communications Unit

Remote

SUMMER INTERNSHIP

Jul.2021 - Aug.2021

SAMSUNG Electronics DS Division Foundry Business Unit

Hwaseong, Gyeonggi-do, S.Korea

SAMSUNG TALENT INTERNSHIP PROGRAM (STIP)

Jul.2020 - Aug.2020

Patent

A Universal Image-Generation Framework for Jailbreaking Large Vision-Language Models by Bypassing Safety Alignment

2025

KR PATENT 10-2025-0157353

- Participated as a Key Developer in Patent Development
- Participated through the Project with ETRI

Method and System for Generating Universal Adversarial Perturbations Using High-Sensitivity Components of Vision Encoders in Large-Scale Vision-Language Models

2025

KR PATENT 10-2025-0194468

- Participated as a Key Developer in Patent Development
- Participated through the Project with IITP

Stone size estimation method

2023

KR PATENT 10-2023-0031846

- Participated as a Key Developer in Patent Development
- Participated through the KAIST EE Externship

Honors & Awards

2026	LG Electronics Paper Award , Summer Annual Conference of IEIE	Jeju, S.Korea
2024	LG Electronics Paper Award (2 papers) , Summer Annual Conference of IEIE	Jeju, S.Korea
2023	Certificate , KAIST Leadership and Volunteer Excellence Award Certificate	Daejeon, S.Korea
2022	Certificate , KAIST CS348 Certificate of Appreciation	Daejeon, S.Korea
2018	Bronze Award , SAMSUNG Humantech Paper Award Biology Division	Seoul, S.Korea
2017	3rd Place , Mathematics Education Joint Academic Conference R&E Poster Presentation Competition	S.Korea
2017	1st Place (Minister of Oceans and Fisheries Award) , Marine Biology Research Competition	Incheon, S.Korea
2017	Bronze Award , Korean Mathematics Competition (KMC)	S.Korea
2017	4th Place , Korean Mathematics Olympiad (KMO)	S.Korea
2017	Award Certificate , President of Korean Earth Science Society	S.Korea

Project

Automatic Evaluation Benchmark Generation from LLM Log Data

Jun. 2025 - Aug. 2030

SAMSUNG ELECTRONICS DS

Development of AI Technology with Robust and Flexible Resilience Against Risk Factors

Jan. 2025 - Dec. 2028

ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE (ETRI)

- Team Leader

Penetration Security Testing of ML Model Vulnerabilities and Defense

Jan. 2025 - Dec. 2027

INSTITUTE FOR INFORMATION & COMMUNICATION TECHNOLOGY PLANNING & EVALUATION (IITP)

Scene Text Recognition with Visual Contexts

Sep. 2024 - Dec. 2024

CENTER FOR SECURITY TECHNOLOGY RESEARCH, KAIST

Bypass Techniques for Identifying Vulnerabilities in CAPTCHA

Mar. 2023 - Dec. 2023

CENTER FOR SECURITY TECHNOLOGY RESEARCH, KAIST

Practical Adversarial Attacks of AI Facial Recognition Technology Using Physical Patterns

Mar. 2023 - Dec. 2023

CENTER FOR SECURITY TECHNOLOGY RESEARCH, KAIST

Presentation

KAIST EE Career Concert 2024

Daejeon, S.Korea

PRESENTER

May. 2024

- This program is for undergraduate students in EE who are struggling to decide on their career path
- Present about my undergraduate years, what led me to pursue graduate studies, and what research I am currently conducting

KAIST EE Career Concert 2022

Daejeon, S.Korea

PRESENTER

Nov. 2022

- Shared experiences at the KAIST EE Externship Program

Extracurricular Activity

KAIST Electrical Engineering TA

Daejeon, S.Korea

MEMBER

Sep. 2023 - Present

- Signals and Systems - Fall 2023
- Introduction to Electronics Design Lab - Fall 2023, Fall 2024, Spring 2025
- Digital Image Processing - Spring 2024
- EE Career Development I - Spring 2025, Spring 2026
- Introduction to Machine Learning - Fall 2025

KAIST Student Counseling Assistant (CA)

Daejeon, S.Korea

MEMBER

Mar. 2024 - Feb. 2025, Sep. 2025 - Present

- The CA (Counseling Assistant) program aims to expand counseling support opportunities for students and graduate students to ensure a smooth academic/school life and career planning.
- Participants work autonomously with peers to enhance their development skills and foster personal growth.

MadCamp 11th Cohort Hosted by KAIST CS

Daejeon, S.Korea

MEMBER

Sep. 2019

- MadCamp is an intensive program in collaboration with KAIST, focused on app, web, and game development for undergraduate students in South Korea.
- Participants work autonomously with peers to enhance their development skills and foster personal growth.

KAIST Electrical Engineering Student Council

Daejeon, S.Korea

VICE DEPARTMENT REPRESENTATIVE

Feb. 2020 - Feb.2022

KAIST Dance Crew Illusion

Daejeon, S.Korea

PRESIDENT

Jan. 2020 - Feb.2022

Awarded the Best Team Title in the Outstanding Freshman

Daejeon, S.Korea

MEMBER

Sep. 2019

Awarded the Outstanding Freshman

Daejeon, S.Korea

MEMBER

Sep. 2019

Vice Class Leader of KAIST Freshman Class 6

Daejeon, S.Korea

VICE CLASS LEADER

Mar. 2019 - Feb. 2020